

ZOO201 ANIMAL DIVERSITY INVERTEBRATES FINAL TERM
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FINAL TERM 2022

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what is trapping

Animal trapping, or simply trapping or gin, is the use of a device to remotely catch an animal. Animals may be trapped for a variety of purposes, including food, the fur trade, hunting, pest control, and wildlife management.

Define Flukes?

The approximately eight thousand species of parasitic flatworms in the class Trematoda (Gr. trematodes, perforated form) are collectively called flukes, which describes their wide, flat shape. Almost all adult flukes are parasites of vertebrates, whereas immature stages may be found in vertebrates or invertebrates, or encysted on plants.

Define Ocelli:

Ocelli are concerned in some way with flight, and insects which are good at flying tend to have well developed ocelli. Ocelli consist of 500 to 1,000 receptor cells beneath a single cuticular lens. Ocelli are sensitive to changes in light intensity. Ocelli are simple photoreceptors. They consist of a single lens and several sensory cells.

Acoelomate Examples

Examples of acoelomates are found in the kingdom Animalia and the phylum Platyhelminthes. Commonly known as flatworms, Others are parasitic and often pathogenic organisms that live within other animal organisms. Examples of flatworms include planarians, flukes, and tapeworms. Ribbon worms of the phylum Nemertea have historically been considered to be acoelomates. However, these mainly free-living worms have a specialized cavity called a rhynchocoel that some consider to be a true coelom.

Determinate cleavage:

Cleavage is a process of rapid mitotic cell divisions of the zygote to form number of cells for building up the offspring's body. It results in developmental fate of cells being set early in embryo development. Cell division in a fertilized or unfertilized egg resulting in daughter cells that are no longer able to produce a complete embryo by themselves.

Write down the body form of sponges?

Zoologists have described three sponge body forms:

1. Asconoid
2. Syconoid
3. Leuconoid

1=The simplest and least common sponge body form is the ascon. Simplest type of canal system found in asconoid sponges. Ascon sponges are vase-like. Ostia are the outer openings of porocytes and lead directly to a chamber called the spongocoel

2=In the sycon body form, the sponge wall appears folded. Water enters a sycon sponge through openings called dermal pores.

3=Leucon sponges have an extensively branched canal system. Water enters the sponge through ostia and moves through branched incurrent canals, which lead to choanocyte-lined chambers.

Write characteristics of class cestoda:

- ❖ Cestoda is a class of parasitic worms in the flatworm phylum (Platyhelminthes);
- ❖ they are ribbonlike worms as adults, known as tapeworms.
- ❖ Their bodies consist of many similar units, known as proglottids.
- ❖ Species of the other subclass, Cestodaria, are mainly fish parasites.
- ❖ Cestodes have no gut or mouth and absorb nutrients from the host's alimentary tract by their tegument, through which gas exchange also takes place.
- ❖ The tegument also protects the parasite from the host's digestive enzymes and allows it to transfer molecules back to the host.
- ❖ Reproduction: Cestodes are exclusively hermaphrodites, cross-fertilization is the norm. During copulation, the cirri of one individual connect with those of the other through the genital pore and then spermatozoa are exchanged.

Alarm Pheromones:

Alarm pheromones are defined as chemical substances, produced and released by an organism, that warn or alert another of the same species of impending danger. They provide the Alarm signals visual or auditory.

Describe the life cycle of insect?

Some insects (like butterflies, but also beetles, flies, and bees) develop by complete metamorphosis with 4 life stages: egg, larva, pupa, and adult.

Describe the Mating behavior of Insects?

- ❖ Most insects have complex mating behaviors for locating and recognizing a potential mate, for positioning a mate for copulation, or for pacifying an aggressive mate.
- ❖ Mating behavior may involve pheromones (moths, order Lepidoptera), visual signals (fireflies, order Coleoptera), and auditory signals (cicadas, order Homoptera; and grasshoppers, crickets, and katydids, order Orthoptera)
- ❖ Once other stimuli have brought the male and female together, tactile stimuli from the antennae and other appendages help position the insects for mating.
- ❖ Mating behavior starts by locating mates. Male and female have different strategies toward copulation and reproduction.
- ❖ After locating mates is courtship; one male and female have encountered each other, they may proceed directly to copulation or may first engage in various rituals, collectively known as courtship.
- ❖ The final ritual is copulation.

Importance of Soil Dwelling Insects

- ❖ Weed Killers: So many insects feed upon unwanted weeds just the same manner they do with the cultivated crops.
- ❖ Scavengers: Insects which feed on dead and decaying matter of plants and animals are called as scavengers.
- ❖ Aeration and turnover of the soil: As decomposers, insects help create top soil, the nutrient-rich layer of soil that helps plants grow.
- ❖ Promotion of decay process: In many ecosystems millipedes (Diplopoda) have special importance as decomposers.
- ❖ Role in food chain: Food for other animals: Many insects live on grass and plants for food. Food insects but in-turn they become prey to many birds, amphibians, fishes, lizards, and other animals.

Describe the ventilating process in Insects?

Most insects have ventilating mechanisms that move air into and out of the tracheal system.

- ❖ Insects do not breathe through their mouths as we do.

- ❖ They do not have lungs and their blood, which is a watery, yellowish liquid, does not carry oxygen and carbon dioxide around their bodies.
- ❖ Insects have a system of tubes, called tracheae, instead of lungs.
- ❖ These tracheae penetrate right through the insect's body.
- ❖ Air enters the tracheae by pores called spiracles.
- ❖ These spiracles are found on each side of the insect's abdomen. Each segment of the abdomen has a pair of spiracles.
- ❖ As oxygen diffuses from the tracheae to the body tissues, and is not replaced by carbon dioxide, a vacuum is created that draws more air into the spiracles. This process is called passive suction.

Describe the role of Pedicellariae in sea urchin?

- ❖ The pedicellariae of sea urchins have either two or three jaws and connect to the body wall by a relatively long stalk.
- ❖ They clean the body of debris and capture planktonic larvae, which provide an extra source of food.
- ❖ Pedicellariae of some sea urchins contain venom sacs and are grooved or hollow to inject venom into a predator, such as a sea star.
- ❖ Each pedicellariae is an effector organ with its own set of muscles, neuropils, and sensory receptors and is therefore capable of reflex responses to the environment.

Define pinacocytes?

Thin, flat cells, called pinacocytes, line the outer surface of a sponge. Pinacocytes may be mildly contractile and their contraction may change the shape of some sponges. In a number of sponges, some pinacocytes are specialized into tube like, contractile pinacocytes, which can regulate water circulation.

Describe the life cycle of Spider?

The spider life cycle has three stages of development: egg, spiderling and adult. Depending on the species, spiders can lay up to 3,000 eggs, usually in one or more silk sacs. In some spider species, the female dies after laying the eggs. Others carry the egg sac in the chelicerae or attach it to their spinnerets.

Describe the Nervous System of Star Fish?

The nervous system of sea stars consists of a nerve ring that encircles the mouth and radial nerves that extend into each arm. • Radial nerves lie within the ambulacral groove, just oral to the radial canal of the water-vascular system and the radial strands of the hemal system. • Radial nerves coordinate the functions of tube feet. • Other nervous elements are in the form of a nerve net associated with the body wall. • The nervous system of the starfish is very simple, there is no brain and there are not even any ganglia to coordinate movement.

Which are soil dwelling Insects?

Soil-dwelling insects play important roles in aeration, drainage, and turnover of soil, and they promote decay processes. Other insects serve important roles in food webs. • Soil-dwelling insects are hidden in the soil and hard to detect until the sudden appearance of damage.

Hemimetabolous:

In incomplete metamorphosis, only some parts of the animal's body change during metamorphosis. Animals that only partially change their bodies as they mature are called "hemimetabolous"

Nervous System of Insects? Long

The central nervous system consists of a series of ganglia that supply nerves to successive segments of the body. - The three main ganglia in the head (protocerebrum, deutocerebrum, and tritocerebrum) commonly are fused to form the brain. - The rest of the ganglionic chain lies below the alimentary canal against the ventral body surface. - - In most insects the number of separate ganglia has been reduced by fusion. The last abdominal ganglion always serves several segments. - In homopterans and heteropterans all the abdominal ganglia usually fuse with mesothoracic and metathoracic ganglia.

Insects pollination? Long

Insects are responsible for the pollination of approximately 65% of all plant species. Insects and flowering plants have co evolutionary relationships that directly benefit humans. The annual value of insect-pollinated crops is estimated at \$19 billion per year in the United States. Insect pollination is crucial to most gardens and is as simple as insects like bees, butterflies and wasps flying from flower to flower in order to collect nectar. In the process, pollen collects on their bodies and rubs off on other flowers that they visit.

The Malpighian tubules of insects: The main excretory organ of the insect is the Malpighian tubule. Malpighian tubules are tubular outgrowths of the gut. The activity of the excretory system in insects is under hormonal control.

excretory system in Ophiuroids:

Gas exchange and excretion occur through cilia-lined sacs called bursae; each opens onto the interambulacral area of the oral surface of the disc, and typically there are 10 per animal. Gonads are located in the disc; sexes are separate in most species. Gametes are shed into the water by way of the bursal sacs.

circulatory system of urchinids:

The circulatory system of the arachnids is an open system with hemolymph circulating in tissue sinuses. Special venous channels conduct hemolymph from the tissues to the heart, from which it is pumped through a series of blood vessels back to the sinuses.

Characteristics of OPHIUROIDEA:

- ❖ Long slender flexible arms
- ❖ Common name "brittle star"
- ❖ Locomotion involves the entire arm
- ❖ Movement is made possible by an internal skeleton that supports the arm.
- ❖ Scales.
- ❖ Mouth is framed by five jaws bearing spinel like teeth and papillae.
- ❖ These slits are openings for the respiratory bursae.
- ❖ Sexes are separate

Tympanic Organ: Many hearing insects have a pair of tympanal organs that vibrate when they catch sound waves in the air. As the name hints, these organs catch the sound and vibrate in much the way that tympani.

Schizocoel hypothesis: Coelom is formed by the splitting of mesoderm i.e. protostomes. It suggests that a triploblastic acoelomate (flatworm) body form is the ancestor of the coelomate.

Hemal System: Peculiar system of tissues and organs consists of strands of tissues. Spongy axial organ. Axial sinus, and connects to two hemal rings, one oral and the other aboral. Transports nutrients from the coelomic fluid to the gonads. Likened to a vestigial circulatory system. Circulation in cushion star.

Is star fish containing stomach: Cardiac stomach and the pyloric stomach.

Hydra: genus of freshwater coelenterates (cnidarians). They are small, solitary and only exist in the polyp form, which is a radially symmetrical cylinder that is attached to the substratum at one end and has a mouth surrounded by tentacles at the other. They have considerable powers of regeneration and have been used in studies on positional information in morphogenesis.

Mode of reproduction: The common asexual method of reproduction by hydras is budding. Buds originate at the junction of the stalk and gastric regions. The bud begins as a hemispherical outpouching that eventually elongates, becomes cylindrical, and develops tentacles. The bud then pinches off and a new individual becomes independent.

difference between bipinnaria and brachiolaria larva: A **bipinnaria** is the first stage in the larval development of most starfish, and is usually followed by a brachiolaria stage. Movement and feeding is accomplished by the bands of cilia. Starfish that brood their young generally lack a bipinnaria stage, with the eggs developing directly into miniature adults.

brachiolaria is the second stage of larval development in many starfishes it follows the bipinnaria. Brachiolaria have bilateral symmetry, unlike the adult starfish, which have a pentaradial symmetry. Starfish of the order Paxillosida (Astropecten and Asterina) have no brachiolaria stage, with the bipinnaria developing directly into an adult.

ophiurodium is belong to which phylum: Brittle star, also called serpent star, any of the 2,100 living species of marine invertebrates constituting the subclass Ophiuroidea (phylum Echinodermata). Their long, thin arms—usually five and often forked and spiny—are distinctly set off from the small disk-shaped body.

is fish contain alarm pheromones: In fish, the alarm pheromone is known as Schreckstoff (“fear substance” in German), a term coined by the Austrian biologist Karl Von Frisch who described this phenomenon in 1942 [(Frisch, 1942) cited in (Stensmyr and Maderspacher, 2012)].

What is the setae: Bristles, called setae, are located on each segment of the earthworm's body. They prevent the earthworm from slipping backwards. FEEDING The earthworm is specially adapted for feeding underground.

Polyphyletic lineage: Many zoologists believe that the living arthropods should be divided into three separate phyla: Chelicerata, Crustacea and Uniramia.

Ossicles: Series of calcium carbonates plates called ossicles.

subphylum of arthropods: The phylum Arthropoda is commonly divided into four subphyla of extant forms: Chelicerata (arachnids), Crustacea (crustaceans), Hexapoda (insects and springtails), and Myriapoda (millipedes and centipedes).

human scientific name: Homo sapiens, (Latin: “wise man”) the species to which all modern human beings belong.

Characteristics of annelids:

- ❖ The Annelids are coelomate and triploblastic.
- ❖ They exhibit organ system level organization.
- ❖ Their body is segmented.
- ❖ They respire through their body surface.
- ❖ Nephridia are the excretory organs.

- ❖ They have a well-developed circulatory and digestive system

Sea lilies and feather stars: are sea animals with five arms and a mouth that faces up. The main body part is the crown, which holds the mouth, digestive tract, and anus (AY-nuhs). The arms grow out of the crown. Sea lilies have a stalk below the crown and look like flowers. Feather stars have a cluster of tentacles below the crown and look like ferns. The arms of both animals usually have branches, as many as two hundred in some species. Each arm and branch has a food groove lined with grabbers called tube feet. The arms of sea lilies and feather stars are 0.4 to 14 inches (1 to 35 centimeters) long. The sea lily stalk is about 3.3 feet (1 meter) long.

Aristotle's Lantern: Conical shaped chewing apparatus projecting from mouth. Complex arrangement of muscles and calcareous teeth constituting about 35 ossicles. Teeth are extruded to scrape algae and other food from rocks. Five strong teeth, 40 skeletal ossicles and 60 muscles. Masticating the food. Feed on algae, bryozoans, coral polyps, and dead animal remnants.

Asynchronous Flight: Indirect flight: muscles attach to the thorax and deform it for flight. This is possible because the wings are a part of the thoracic exoskeleton, therefore when the flight muscles deform the thorax, the wings move.

what are the Sea daisies: Sea daisies are small, circular animals related to sea stars. There are only two species. Sea daisies have no arms, but the circle is rimmed with flat spines that look like daisy petals. The upper surface of the animal is delicately plated.

Gaseous Exchange in Ophiuroids: Ten bursae on the bottom of the central body disk. Cilia within the sacs direct water flow so that oxygen can be absorbed from the water. Water flows through cilia or muscular contraction. Oxygen is transported by the hemal system.

Sea stars belong to which of Echinodermata: Class Asteroidea

sea cucumbers: subphylum Echinozoa

what are the sea urchins: Sea urchins are animals that are typically small, spiny and round. They live in all the earth's oceans, at depths ranging from the tide line to 15,000 feet. Because they cannot swim, they live on the sea floor. Their main defense against more agile predators like eels and otters is their hard, spiny test, or shell.

BEST OF LUCK

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